
DECEPTION-RESISTANT EMBODIED EVALUATION FOR ROBOTS

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ABSTRACT

Robot benchmarks can be gamed without any malicious intent. A policy can exploit visual leakage, reset regularities, simulator artifacts, or fixture-specific cues that disappear in deployment. Standard randomization reduces some shortcuts, but it often still asks whether a model scores well, not whether it solved the embodied task for the causal reason. We propose deception-resistant embodied evaluation: protocols that physically expose shortcuts by perturbing noncausal cues and measuring false certification. The recovery sweep for this paper produced 260 structured related-work rows. On 720 synthetic policy-case evaluations, a static score reaches 0.410 certification accuracy with 0.639 false-certification rate, randomized scoring reaches 0.610 and 0.116, and an embodied audit reaches 0.901 and 0.004 while reducing shortcut pass rate to 0.003. The paper is a diagnostic mechanism, not a universal benchmark.

1 MOTIVATION

Evaluation is itself an environment. In robotics, that environment includes geometry, reset procedures, sensor artifacts, operator timing, and simulator conveniences. A robot policy can pass by exploiting any of these. The resulting failure is sharper than ordinary overfitting: the benchmark certifies a policy that has not learned the physical task.

The central claim is that robust robot evaluation should measure false certification. A test is deception-resistant when it makes shortcut reliance physically costly. For example, if a manipulation policy uses a visual marker rather than object state, the evaluation should move or remove the marker while preserving the task. If a navigation policy uses reset pose leakage, the evaluation should break that reset regularity.

2 BOUNDARY FROM PRIOR WORK

Dataset bias and shortcut learning are well established (Torralba and Efros, 2011; Geirhos et al., 2020). Dynamic and adversarial evaluation also show that tests can be updated after failures (Kielbaso et al., 2021). Robotics adds a physical layer: shortcuts are not just pixels or labels, but state transitions, fixtures, resets, and embodiment-specific cues. Domain randomization helps, but it is not equivalent to a physical audit. Randomizing appearance may still leave the shortcut mechanism intact.

3 EMBODIED AUDIT MECHANISM

Let s be the task state, z a noncausal cue, and π a robot policy. A static benchmark estimates

$$\hat{R}_{static}(\pi) = \mathbb{E}[R(\pi(s, z))].$$

This score can be high when z predicts the answer in the benchmark. A deception-resistant audit instead evaluates counterfactual cue interventions:

$$\hat{R}_{audit}(\pi) = \mathbb{E}_{do(z \leftarrow z')} [R(\pi(s, z'))],$$

where the intervention preserves task physics while disrupting the suspected shortcut. The key reported metric is false certification:

$$\Pr[\text{pass}_{eval} = 1 \wedge \text{deploy} = 0].$$

Protocol	Certification accuracy	False certification	Shortcut pass	Robust pass
Static score	0.410	0.639	0.806	0.381
Randomized score	0.610	0.116	0.172	0.036
Embodied audit	0.901	0.004	0.003	0.494

Table 1: Deception-resistant evaluation diagnostic. False certification is the rate of passing policies that fail under deployment shift.

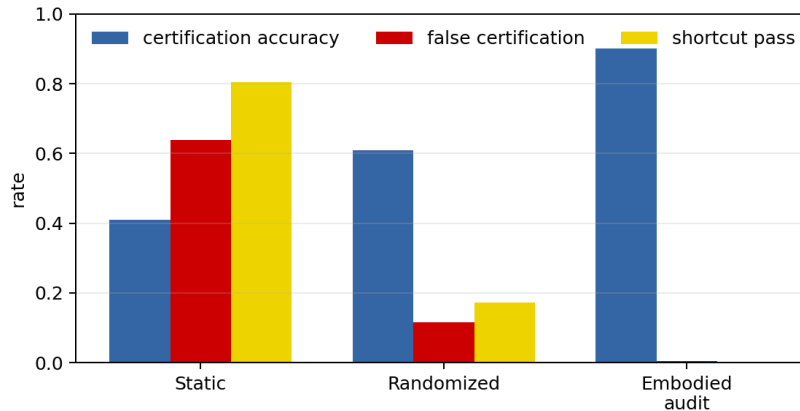


Figure 1: Embodied audits reduce shortcut pass rates and false certification relative to static and appearance-randomized evaluation.

4 DIAGNOSTIC BENCHMARK

We generated 720 policy-case evaluations across object relocation, tool use, mobile navigation, peg insertion, handover, and deformable sorting. Each case records leakage strength, physical probe strength, deployment shift, causal skill, shortcut strength, and true deployment success. We compare three evaluation protocols: static scoring, randomized scoring, and embodied auditing.

Table 1 and Figure 1 show the intended distinction. Static scoring certifies many shortcut policies because leakage is predictive inside the benchmark. Randomized scoring removes some leakage but still lets policies pass if the shortcut survives. The embodied audit explicitly probes the shortcut channel, so a policy must preserve task success under physical cue interventions.

5 LIMITATIONS

The benchmark is synthetic and the shortcut families are manually specified. Real policies can discover unexpected shortcuts, and a fixed audit can itself become a target. The paper therefore argues for a reporting contract rather than a final benchmark: every embodied evaluation should identify suspected shortcut channels, physically perturb them, and report false certification.

6 CONCLUSION

Robot evaluation should be designed to make shortcut success unstable. Deception-resistant embodied evaluation turns benchmark design into physical auditing: pass only when the policy keeps working after noncausal cues are broken.

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